Notes: Donangelo (JF, 2014)
Labor Mobility: Implications for Asset Pricing

Contents

1 One-sentence summary

Labor mobility raises shareholders' exposure to systematic risk because mobile workers leave or demand outside wages when industry opportunities worsen, making cash flows more procyclical and generating a return premium for firms in high-mobility industries.

2 Big picture

2.1 Research question

The paper asks whether firms employing workers with more portable skills face greater systematic risk and therefore higher expected stock returns. The mechanism is that firms rent labor from workers who can walk away; the less industry-specific the workers' skills are, the more responsive the firm's effective labor supply is to outside opportunities and shocks.

2.2 Main asset-pricing claim

Industries with high labor mobility behave as if they have higher operating leverage. When industry conditions deteriorate, mobile workers can leave or wages cannot fully absorb the shock, making shareholders absorb more of the downside. Investors therefore require higher expected returns for holding firms in mobile industries.

2.3 Core empirical result

A portfolio long high-labor-mobility stocks and short low-labor-mobility stocks earns an economically large return spread. In the portfolio sorts, the high-minus-low labor-mobility spread is about 0.5% per month for several return measures, or more than 5% per year.

2.4 Mechanism in one sentence

Labor mobility makes wages less sensitive and profits more sensitive to industry shocks; this creates labor-induced operating leverage.

2.5 Why the paper matters

The paper connects labor-market characteristics to asset prices. It shows that the organization of the workforce - not just technology, capital intensity, or financial leverage - can explain differences in firms' risk premia.

3 Incremental contribution of the paper

3.1 A labor-market source of systematic risk

The paper adds a new primitive to asset pricing: workers' outside options. Existing operating leverage theories often emphasize fixed costs, capital intensity, or financial leverage. Donangelo shows that labor mobility itself can be a source of operating leverage because labor is not owned by firms.

3.2 A new empirical measure of labor mobility

The paper constructs an industry-level measure of labor mobility from occupation dispersion across industries. Occupations concentrated in few industries are interpreted as requiring more industry-specific human capital and hence low mobility; occupations spread across many industries are interpreted as more mobile.

3.3 A bridge between labor economics and asset pricing

Labor economics studies occupation-specific and industry-specific human capital; asset pricing studies cross-sectional risk premia. This paper links the two by showing that occupational portability predicts both real sensitivities to shocks and stock returns.

3.4 Mechanism evidence before return tests

The paper does not simply sort stocks on a characteristic. It first shows that mobility predicts employment elasticity, wage elasticity, and profit elasticity with respect to industry shocks. This links the return premium to a cash-flow risk mechanism.

3.5 Multiple expected-return proxies

The paper tests realized portfolio returns, panel return regressions, implied cost of capital measures, and CAPM betas. This helps show that the labor-mobility premium is not an artifact of one return metric.

4 Data and measurement

4.1 Industry-level labor mobility measure

The paper builds a measure of labor mobility using the BLS occupation-by-industry distribution. The idea is to infer how portable skills are from where occupations appear in the economy. If an occupation is concentrated in one industry, its skills are likely industry-specific. If an occupation appears in many industries, its workers have more outside options.

At the occupation level, the paper constructs a concentration measure based on employment shares across industries. At the industry level, labor mobility is the wage-weighted average mobility of the occupations used by that industry. The measure is standardized cross-sectionally each year.

Interpretation: The measure is indirect but economically transparent. It does not require observing worker moves after shocks. Instead, it uses the cross-industry dispersion of occupations as a proxy for the portability of worker skills.

4.2 Firm and industry data

The asset-pricing tests merge the labor-mobility measure with CRSP and Compustat. Firm-level variables include size, book-to-market, leverage, profitability, organization capital, market beta, labor intensity, and other controls. The main return sample covers roughly 1991 to 2012.

4.3 Labor-related variables

The paper uses wage and employment data from BLS/OES and BLS/KLEMS. Key labor-side objects include employment growth, wage growth, industry labor intensity, and the sensitivity of profits/wages/employment to productivity shocks.

4.4 Expected-return proxies

The paper studies expected returns in four ways:

- sorted portfolio returns by labor mobility;
- panel regressions of annual stock returns on lagged mobility and controls;
- implied cost of capital measures from Gebhardt-Lee-Swaminathan, Wu-Zhang, and Hou-van Dijk-Zhang;
- unconditional and conditional CAPM betas.

5 Model setup and theoretical mechanism

5.1 Economic environment

The model features firms in an industry competing for workers in a partially segmented labor market. The key industry characteristic is how easily workers can move to alternative industries. Workers with general skills have stronger outside options, so the industry faces a more elastic labor supply.

5.2 Production and shocks

Firms produce output using capital and labor. Industry productivity shocks change the marginal product of labor and capital. When workers are mobile, they can leave after unfavorable shocks, making employment and cash flows more sensitive to shocks.

5.3 Labor-induced operating leverage

Traditional operating leverage comes from fixed costs. Labor mobility creates an analogous effect because firms cannot fully adjust the unit labor cost of mobile workers. In downturns, mobile workers' outside options and flows amplify the losses borne by shareholders.

5.4 Predicted real elasticities

The model predicts:

- high-mobility industries have more profit sensitivity to productivity shocks;
- high-mobility industries have less wage sensitivity to productivity shocks;
- employment responses can be larger or differently timed in high-mobility industries;
- these real-side elasticities translate into higher systematic risk for equity owners.

5.5 Expected returns

The asset-pricing implication is:

$$E[R_H - R_L] > 0,$$

where H denotes firms in high-mobility industries and L denotes firms in low-mobility industries. The return spread compensates investors for the extra systematic risk induced by mobile labor.

6 Empirical specification and identification design

6.1 Occupation-dispersion design

The key measurement assumption is that occupational dispersion across industries captures the portability of skills. For example, airline pilots are concentrated in air transportation and thus have low cross-industry mobility; managers, clerks, and accountants appear across many industries and thus are more mobile.

6.2 Industry-shock regressions

The paper estimates time-series regressions of industry outcomes on productivity shocks:

$$\Delta Y_{j,t} = \alpha_j + \beta_j \Delta \mathrm{TFP}_{j,t} + \varepsilon_{j,t},$$

where Y is profitability, wages, or employment. It then compares average slopes across high- and low-mobility industries.

Interpretation: This test asks whether the real side of the economy behaves as the theory predicts. If mobility creates operating leverage, profit sensitivity to industry shocks should be higher in high-mobility industries.

6.3 Portfolio-sorting design

Firms are sorted into labor-mobility portfolios using lagged mobility. The portfolio analysis compares high- versus low-mobility stocks in value-weighted and equal-weighted portfolios, and uses raw, adjusted, excess, and unlevered returns.

6.4 Panel return regressions

The panel specification is:

$$R_{i,t} = \alpha_t + \beta \mathrm{LM}_{i,t-1} + \gamma' X_{i,t-1} + \varepsilon_{i,t},$$

where X contains known return predictors such as size, book-to-market, lagged returns, market beta, labor intensity, and leverage.

6.5 Implied cost of capital tests

The paper also estimates:

$$\mathrm{ICC}_{i,t} = \alpha_t + \beta \mathrm{LM}_{i,t-1} + \gamma' X_{i,t-1} + \varepsilon_{i,t}.$$

Using implied cost of capital reduces dependence on realized return noise and provides an alternative test of the pricing implication.

6.6 CAPM beta tests

The final return-side test examines whether high labor mobility is associated with higher market betas. The paper estimates unconditional and conditional CAPM models for labor-mobility-sorted portfolios.

6.7 Identification interpretation

The paper is not a natural experiment. Identification comes from combining a model-implied measurement strategy with multiple cross-sectional and time-series tests. The strongest interpretation is that the same mobility measure predicts real shock sensitivities and expected returns, which is exactly what the model predicts.

6.8 Limitations

The labor-mobility measure is a proxy, not direct worker mobility. It may be correlated with industry characteristics such as manufacturing intensity, unionization, labor intensity, or technology. The paper addresses this with controls, alternative expected-return measures, and mechanism tests, but the design remains observational.

7 Tables and figures

7.1 Table I: Most Immobile and Most Mobile Occupations; Table II: Most Immobile and Most Mobile Industries

Read: Table I ranks occupations by labor mobility. Highly immobile occupations include postmasters, embalmers, funeral attendants, subway operators, and dental hygienists. Highly mobile occupations include general and operations managers, planning clerks, marketing managers, production supervisors, and bookkeeping/accounting clerks. Table II aggregates this logic to industries, showing low mobility in rail transportation, hospitals, colleges, mining, and broadcasting, and high mobility in several manufacturing and wholesale industries.

Economic message: The tables validate the mobility measure. Occupations and industries that intuitively require specialized skills receive low mobility scores; broad generalist occupations and industries receive high mobility scores.

Table I

Most Immobile and Most Mobile Occupations

This table presents the bottom 15 (Panel A) and top 15 (Panel B) five-digit SOC broad occupations sorted on the measure of labor mobility, as of 2011. The ranking excludes occupations that do not require at least a few months of specific preparation (occupations with Specific Vocational Preparation below four from the DOT).

SOC	Occupation Title	Mobility
	Panel A: Bottom 15 Occupations by Labor Mobility	
119130	Postmasters and Mail Superintendents	0.01
394010	Embalmers	0.01
394020	Funeral Attendants	0.01
534040	Subway and Streetcar Operators	0.02
372020	Building Cleaning Workers, All Other	0.03
537110	Mine Shuttle Car Operators	0.04
292020	Dental Hygienists	0.06
519140	Semiconductor Processors	0.06
474020	Elevator Installers and Repairers	0.08
475060	Roof Bolters, Mining	0.08
291010	Chiropractors	0.09
534020	Railroad Brake, Signal, and Switch Operators	0.09
413040	Travel Agents	0.09
331020	First-Line Supervisors of Fire Fighting and Prevention Workers	0.12
474060	Rail-Track Laying and Maintenance Equipment Operators	0.12
	Panel B: Top 15 Occupations by Labor Mobility	
111020	General and Operations Managers	82.89
435060	Production, Planning, and Expediting Clerks	72.86
112020	Marketing Managers	64.79
511010	First-Line Supervisors of Production and Operating Workers	59.75
433030	Bookkeeping, Accounting, and Auditing Clerks	56.05
499040	Industrial Machinery Mechanics	51.49
436010	Executive Secretaries and Executive Administrative Assistants	51.25
113050	Industrial Production Managers	48.16
173020	Drafters, All Other	45.36
435070	Shipping, Receiving, and Traffic Clerks	45.27
131020	Buyers and Purchasing Agents, Farm Products	45.25
439060	Office Clerks, General	43.66
519060	Inspectors, Testers, Sorters, Samplers, and Weighter	41.25
514030	Cutting, Punching, and Press Machine Operators	40.79
111010	Chief Executives	39.60

(a) Table I: Most immobile and most mobile occupations.

Table II

Most Immobile and Most Mobile Industries

This table presents the bottom 15 (Panel A) and top 15 (Panel B) four-digit NAICS industries represented in the merged CRSP/Compustat data set, sorted on the measure of labor mobility, as of 2011.

SOC	Occupation Title	Mobility
	Panel A: Bottom 15 Industries by Labor Mobility	
482100	Rail Transportation	−1.71
812200	Death Care Services	−1.70
561500	Travel Arrangement and Reservation Services	−1.59
238200	Building Equipment Contractors	−1.50
481100	Scheduled Air Transportation	−1.44
481200	Nonscheduled Air Transportation	−1.35
611200	Junior Colleges	−1.31
622100	General Medical and Surgical Hospitals	−1.27
611300	Colleges, Universities, and Professional Schools	−1.26
212100	Coal Mining	−1.24
621600	Home Health Care Services	−1.20
213100	Support Activities for Mining	−1.18
721100	Traveler Accommodation	−1.16
541300	Architectural, Engineering, and Related Services	−1.08
515100	Radio and Television Broadcasting	−1.08
	Panel B: Top 15 Industries by Labor Mobility	
315200	Cut and Sew Apparel Manufacturing	2.29
447100	Gasoline Stations	1.99
423500	Metal and Mineral (except Petroleum) Merchant Wholesalers	1.94
425100	Wholesale Electronic Markets and Agents and Brokers	1.77
424600	Chemical and Allied Products Merchant Wholesalers	1.76
532200	Consumer Goods Rental	1.60
312200	Tobacco Manufacturing	1.57
332800	Metal Heat Treating	1.55
325500	Paint, Coating, and Adhesive Manufacturing	1.48
424400	Grocery and Related Product Merchant Wholesalers	1.44
325600	Soap, Cleaning Compound, and Toilet Preparation Manufacturing	1.44
335100	Electric Lighting Equipment Manufacturing	1.44
334300	Audio and Video Equipment Manufacturing	1.43
314100	Textile Furnishings Mills	1.41
423600	Electrical and Electronic Goods Merchant Wholesalers	1.39

(b) Table II: Most immobile and most mobile industries.

The first pair of tables defines the paper's core measurement object. Table I constructs intuition at the occupation level; Table II shows how that occupation-level measure aggregates to industries used in the return tests.

7.2 Table III: Summary Statistics; Table IV: Labor Mobility and Sensitivity of Profits, Wages, and Employment to Industry Shocks

Read: Table III shows that high-mobility firms are more labor intensive, have somewhat different labor characteristics, are smaller, and have higher market betas than low-mobility firms. Table IV shows that high-mobility industries have higher profit sensitivity and lower wage sensitivity to productivity shocks.

Economic message: These tables supply the mechanism evidence. High-mobility firms are not just different on paper; their cash flows respond more strongly to industry shocks, consistent with labor-induced operating leverage.

Table III

Summary Statistics

This table reports time-series averages of median characteristics of portfolios of firms sorted by labor mobility, LM. Variables are from CRSP/Compustat, unless otherwise noted. *Lab. Int.* is the logarithm of the ratio of the number of employees divided by PPE and *L. Emp.* is the logarithm of the number of employees. *% Skill* is the fraction of workers in occupations with Specific Vocational Preparation (SVP) above six from the Dictionary of Occupational Titles (DOT). *Occ. Prep.* is constructed as the JobZone index from O*NET. *Union* is the percentage of employees covered by union membership, based on data from unionstats.com. *L. Wage* is the logarithm of annual wages per worker from the BLS. σ_{wg} is the standard deviation of annual wage growth from OES/BLS. $\rho_{wg,gdp}$ is the correlation of annual wage growth with GDP growth (from the St. Louis Federal Reserve). *L.Size* is the logarithm of the market value of equity plus book value of total debt. *L.Asset* is the logarithm of the book value of assets. *B/M* is shareholders equity divided by the market value of equity. *Lev.* is the ratio of the book value of debt adjusted for cash holdings to total assets. *Profit* is the ratio of EBITDA to revenues. *Op. Lev.* is operating leverage, defined as the slope of a time-series regression of payments to a unit of capital growth on TFP growth, using data from KLEMS/BLS. *MKT Beta* is constructed each year with monthly equally weighted portfolio returns. *HHI* is the Herfindahl-Hirschman Index of industry asset concentration, constructed as in Hou and Robinson (2006). *OC* is the measure of organization capital from Eisfeldt and Papanikolaou (2013). The sample covers the period 1990 to 2011, except where otherwise noted.

Panel A: Labor Related Characteristics

LM Portfolio	LM	Lab. Int.	L. Emp	% Skill.	Occ. Prep.	Union	L. Wage*	σ_{wg}*	$\rho_{wg,gdp}$*
L	–0.94	3.03	7.25	36.8	3.07	0.10	2.92	2.15	0.09
2	–0.19	3.49	7.18	40.1	3.27	0.09	3.10	2.17	0.13
3	0.23	3.53	6.36	48.1	3.37	0.11	3.12	3.47	–0.04
4	0.88	3.66	6.48	48.0	3.39	0.08	3.19	2.39	0.04
H	1.30	3.66	6.65	43.6	3.00	0.11	2.96	1.89	0.14

Panel B: Financial and Accounting Characteristics

LM Portfolio	L. Size	L. Asset	B/M	Lev.	Profit	Op. Lev.	MKT Beta	HHI	OC
L	5.42	5.69	0.57	0.37	0.13	1.16	1.33	0.16	0.32
2	5.37	5.42	0.54	0.30	0.08	3.63	1.57	0.13	0.71
3	5.16	4.88	0.44	0.23	0.07	3.11	1.61	0.11	0.80
4	5.09	4.89	0.49	0.25	0.10	4.28	1.63	0.15	0.83
H	5.08	5.02	0.50	0.27	0.09	4.26	1.67	0.19	0.86

(a) Table III: Summary statistics by labor-mobility portfolio.

Table IV

Labor Mobility and Sensitivity of Profits, Wages, and Employment to Industry Shocks

The table reports the average slope of univariate time-series regressions of percentage changes in profitability and labor-related variables on percentage changes in total factor productivity. Data are from the Manufacturing and Nonmanufacturing KLEMS/BLS and from the OES/BLS data sets. A list of the broad industry groups used in the KLEMS/BLS data set is provided in the Internet Appendix. Industry groups are assigned to labor mobility subsamples based on their average cross-sectional ranking. Δ*Profits* is growth in the ratio of *payments to capital* over *productive capital stock*. Δ*Wages* is growth in *price of labor*, that is, compensation per employee-hour worked. Δ*Emp* is growth in number of employees in the broad industry group. Δ*KLEMS TFP* is multifactor productivity growth. Δ*Adjusted TFP* is the residual of time-series regressions of Δ*KLEMS TFP* growth on lagged employment and productive capital growth. Huber–White robust standard errors are shown in parentheses. Significance levels are denoted by * = 10% level, ** = 5% level, and *** = 1% level. The sample covers the period 1990 to 2010.

		Dependent Variable			
Sample	Industries	ΔProfits$_t$	ΔWages$_t$	ΔEmp$_t$	ΔEmp$_{t+1}$
Panel A: Sensitivities to ΔKLEMS TFP$_t$					
Full sample	57	2.23***	0.68***	–0.09	0.00
		(0.38)	(0.13)	(0.05)	(0.06)
Low mobility	28	1.22***	0.94***	–0.10	–0.10
		(0.36)	(0.25)	(0.07)	(0.07)
High mobility	29	3.21***	0.44***	–0.09	0.09
		(0.64)	(0.08)	(0.08)	(0.10)
H-L (*t*-test)		2.00***	–0.50*	0.01	0.19*
		(0.73)	(0.26)	(0.10)	(0.10)
Panel B: Sensitivities to ΔAdjusted TFP$_t$					
Full sample	57	2.31***	0.68***	0.07	0.05
		(0.38)	(0.13)	(0.04)	(0.06)
Low mobility	28	1.34***	0.99***	0.02	–0.08
		(0.37)	(0.25)	(0.05)	(0.07)
High mobility	29	3.25***	0.38***	0.12*	0.18*
		(0.61)	(0.07)	(0.07)	(0.10)
H-L (*t*-test)		1.92***	–0.61**	0.11	0.26**
		(0.71)	(0.26)	(0.09)	(0.12)

(b) Table IV: Sensitivities to industry shocks.

Table III describes what high- and low-mobility firms look like; Table IV tests the central real-side mechanism linking labor mobility to operating leverage.

7.3 Table V: Cross-Section of Returns of Stocks Sorted on Labor Mobility; Table VI: Panel Data Regressions of Annual Stock Returns on Labor Mobility and Firm Characteristics

Read: Table V shows that high-mobility portfolios earn higher average returns than low-mobility portfolios. The high-minus-low spread is economically large and remains positive across raw, adjusted, excess, and unlevered returns. Table VI confirms the result in panel regressions: lagged mobility positively predicts annual stock returns after controlling for firm characteristics.

Economic message: These tables are the main asset-pricing evidence. The mobility premium is not only visible in portfolio sorts; it survives regression controls for size, value, past returns, market beta, labor intensity, and leverage.

Table V

Cross-Section of Returns of Stocks Sorted on Labor Mobility

This table reports post-ranking mean realized excess monthly stock returns over one-month Treasury bill rates, and adjusted monthly returns of portfolios of stocks sorted on labor mobility. Labor mobility is lagged 18 months. *Excess* are returns minus the one-month Treasury bill and *Unlevered* are estimated as excess stock returns times one minus the lagged leverage ratio (measured using the book value of debt and market value of equity). *Raw* are unadjusted excess returns and *Adj* are returns adjusted for size, book-to-market, and momentum, according to the methodology in Daniel et al. (1997). H-L is the zero investment portfolio long the portfolio of firms with high labor mobility (H) and short the portfolio of firms with low labor mobility (L). Newey–West standard errors estimated with one lag are shown in parentheses. Significance levels are denoted by ** = 5% level and *** = 1% level. The sample covers the period 1991 to 2012.

	Value Weighted				Equally Weighted			
	Excess		Unlevered		Excess		Unlevered	
Portfolio	Raw	Adj.	Raw	Adj.	Raw	Adj.	Raw	Adj.
L	0.58	–0.15	0.30	–0.09	0.84	–0.16	0.39	–0.10
2	0.46	–0.23	0.23	–0.15	0.86	–0.10	0.39	–0.06
3	0.65	–0.03	0.46	0.00	1.06	0.12	0.65	0.16
4	0.65	–0.06	0.37	–0.04	1.13	0.15	0.67	0.15
H	1.12	0.38	0.81	0.34	1.34	0.29	0.80	0.25
H-L	0.54**	0.53***	0.50***	0.42***	0.50**	0.46**	0.41**	0.35**
	(0.22)	(0.18)	(0.17)	(0.13)	(0.25)	(0.19)	(0.20)	(0.14)

(a) Table V: Portfolio returns sorted on labor mobility.

Table VI

Panel Data Regressions of Annual Stock Returns on Labor Mobility and Firm Characteristics

This table shows estimates and standard errors of panel data regressions with year effects of stock returns on labor mobility and firm characteristics. Labor mobility is lagged 18 months. Variables are described in Table III. Standard errors clustered by firm are shown in parentheses. Significance levels are denoted ** = 5% level and *** = 1% level. The sample covers the period 1991 to 2012.

	I	II	III	IV	V	VI	VII	VIII	IX	X
Mobility$_{t-1}$	2.52***	2.64***	2.50***	2.94***	2.76***	2.17***	2.83***	2.97***		
	(0.34)	(0.30)	(0.30)	(0.30)	(0.32)	(0.34)	(0.30)	(0.30)		
Log Size$_{t-1}$	–0.41**		–1.16***							
	(0.18)		(0.14)							
Log B/M$_{t-1}$	4.28***			5.54***					4.57***	
	(0.52)			(0.33)					(0.42)	
Lag Ret.$_{t-1}$	–0.03***				–0.05***				–0.03***	
	(0.00)				(0.00)				(0.00)	
MKT Beta$_{t-1}$	2.30**					2.65***			3.38***	3.17***
	(1.12)					(1.02)			(1.04)	(1.03)
Lab.Int.$_{t-1}$	–0.53**						–0.63***			
	(0.22)						(0.18)			
Leverage$_{t-1}$	0.19							10.82***		
	(1.85)							(1.29)		
Year Eff.	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
R^2(%)	6.67	4.75	4.85	5.23	5.81	6.03	4.76	4.87	6.54	5.95
Obs.	41,833	63,798	63,798	63,798	53,850	42,067	63,798	63,798	41,833	42,067

(b) Table VI: Annual returns on lagged labor mobility and controls.

Table V gives the simple portfolio evidence; Table VI shows that the same premium survives in firm-level panel regressions.

7.4 Table VII: Panel Data Regressions of Implied Cost of Capital on Labor Mobility; Table VIII: Labor Mobility and CAPM Betas

Read: Table VII shows that labor mobility is positively related to multiple implied cost of capital measures. Table VIII shows that high-mobility portfolios have larger CAPM betas, and the unconditional CAPM does not fully price the high-minus-low portfolio.

Economic message: The expected-return result is robust to alternative proxies. Mobility predicts both realized returns and forward-looking cost of capital. It also lines up with greater market risk exposure.

Table VII

Panel Data Regressions of Implied Cost of Capital on Labor Mobility

This table reports estimates and standard errors of panel data regressions with year effects of measures of cost of capital on lagged labor mobility and firm characteristics. *GLS* is the cost of capital measure in Gebhardt, Lee, and Swaminathan (2002). *WZ* is the cost of capital measure of Wu and Zhang (2008). *HVDZ* is the cost of capital measure of Hou, van Dijk, and Zhang (2012). The remaining variables are defined in Table III. Standard errors clustered by firm are shown in parentheses. Significance levels are denoted by * = 10% level, ** = 5% level, and *** = 1% level. The sample covers the period 1991 to 2010.

	GLS$_t$			WZ$_t$			HVDZ$_t$		
Dep. Var.	I	II	III	I	II	III	I	II	III
Mobility$_{t-1}$	0.16***	0.12***	0.08*	0.17***	0.11***	0.03	0.18***	0.13**	0.18***
	(0.04)	(0.04)	(0.05)	(0.03)	(0.03)	(0.05)	(0.05)	(0.05)	(0.07)
Log Size$_{t-1}$	–0.22***	–0.23***		0.13***	0.22***		–0.85***	–0.74***	
	(0.02)	(0.03)		(0.02)	(0.02)		(0.04)	(0.03)	
Log B/M$_{t-1}$	0.79***	1.35***		2.05***	2.44***		1.40***	1.72***	
	(0.08)	(0.07)		(0.06)	(0.06)		(0.09)	(0.08)	
MKT Beta$_{t-1}$	0.45***			–0.87***			–1.10***		
	(0.10)			(0.07)			(0.13)		
Leverage$_{t-1}$	3.43***			2.10***			1.54***		
	(0.22)			(0.17)			(0.29)		
Year Eff.	Y	Y	Y	Y	Y	Y	Y	Y	Y
R^2(%)	9.7	8.1	1.0	28.4	27.0	3.6	19.2	18.6	3.7
Obs.	26,404	26,404	26,404	30,055	30,055	30,055	30,002	30,002	30,002

(a) Table VII: Implied cost of capital on labor mobility.

Table VIII

Labor Mobility and CAPM Betas

This table reports estimates of the unconditional and conditional CAPM using five portfolios of stocks sorted on labor mobility. The table reports the intercept (monthly alpha) and slopes (betas) of time-series regressions of excess portfolio returns on excess market returns. Labor mobility is lagged 18 months. Estimated betas are corrected for nonsynchronous trading following the methodology described in Lewellen and Nagel (2006). Returns are at a monthly frequency. Conditional betas are estimated each calendar year. Newey–West standard errors estimated with one lag are shown in parentheses. Significance levels are denoted by * = 10% level, ** = 5% level, and *** = 1% level. The sample covers the period 1991 to 2012.

	Value-Weighted Portfolios						Equally Weighted Portfolios					
	L	2	3	4	H	H-L	L	2	3	4	H	H-L
Panel A: Excess Returns												
$E[R]-r_f$(%)	0.58*	0.46	0.65**	0.65*	1.12***	0.54**	0.84**	0.86*	1.06**	1.13**	1.34***	0.50**
	(0.35)	(0.40)	(0.28)	(0.34)	(0.31)	(0.22)	(0.40)	(0.48)	(0.44)	(0.44)	(0.47)	(0.25)
Panel B: Unconditional CAPM												
Alpha (%)	–0.06	–0.24	0.20	0.05	0.57***	0.63***	0.12	–0.01	0.30	0.34	0.54**	0.42*
	(0.16)	(0.19)	(0.15)	(0.14)	(0.15)	(0.23)	(0.20)	(0.27)	(0.23)	(0.22)	(0.25)	(0.22)
MKT Beta	1.12***	1.24***	0.80***	1.06***	0.96***	–0.16	1.29***	1.55***	1.35***	1.41***	1.43***	0.13*
	(0.05)	(0.06)	(0.04)	(0.04)	(0.04)	(0.07)	(0.06)	(0.08)	(0.07)	(0.06)	(0.07)	(0.06)
R^2(%)	77.0	77.4	74.4	79.9	78.1	2.5	70.5	67.2	67.6	70.8	66.7	3.1
Panel C: Conditional CAPM												
Avg. Alpha (%)	–0.02	–0.06	0.20*	–0.04	0.32*	0.34	0.07	0.15	0.00	0.21	0.22	0.15
	(0.13)	(0.10)	(0.11)	(0.13)	(0.16)	(0.22)	(0.27)	(0.30)	(0.42)	(0.32)	(0.35)	(0.28)
Avg. MKT Beta	1.03***	1.12***	0.81***	1.16***	1.07***	0.04	1.35***	1.58***	1.67***	1.63***	1.68***	0.32*
	(0.07)	(0.08)	(0.06)	(0.07)	(0.07)	(0.11)	(0.08)	(0.14)	(0.15)	(0.13)	(0.13)	(0.17)
Avg. R^2(%)	81.2	87.0	76.2	83.1	83.3	24.7	72.3	74.8	75.5	74.8	76.5	31.1

(b) Table VIII: Labor mobility and CAPM betas.

Table VII uses forward-looking expected-return measures; Table VIII connects the mobility premium to systematic risk loadings.

8 Short summary

- The paper studies labor mobility as a source of operating leverage and systematic risk.
- Mobility is measured from the dispersion of occupations across industries.

- High-mobility industries rely more on generalist workers with better outside options.
- High-mobility industries have more sensitive profits and less sensitive wages in response to productivity shocks.
- Firms in high-mobility industries earn higher returns than firms in low-mobility industries.
- The premium survives controls for size, value, momentum, beta, labor intensity, and leverage.
- Implied cost of capital and CAPM beta tests reinforce the return evidence.
- The paper links labor-market structure to asset prices through cash-flow risk.

9 Conclusion

9.1 Core conclusion

Donangelo shows that labor mobility is an economically important source of equity risk. Firms in industries that employ mobile workers face more volatile shareholder cash flows because workers with portable skills can move in response to better opportunities. This increases the effective operating leverage borne by capital owners and raises expected stock returns.

9.2 Incremental contribution revisited

The paper's main contribution is to identify a new, measurable source of systematic risk: the portability of workers' skills. It creates an empirical measure of labor mobility, validates it using labor-market and cash-flow elasticities, and shows that the same measure prices stocks.

9.3 Economic intuition

Capital is relatively stuck inside the firm, while labor can leave. When workers are more mobile, they absorb less of the downside risk through wages and employment adjustment; shareholders absorb more. Investors require compensation for holding firms with this labor-induced exposure.

9.4 Implications for asset pricing

The results imply that firm-level or industry-level workforce characteristics can be priced risk factors. Labor-market frictions, human-capital specificity, and occupational composition are not only labor-economics variables; they help determine expected equity returns.

9.5 Limitations and open questions

The measure of labor mobility is indirect and industry-level. It may capture correlated industry features, though the paper controls for many firm characteristics. Future work could use worker-level flows, geographic mobility, occupation transitions, or matched employer-employee data to refine the mechanism.

9.6 Takeaway

The paper's takeaway is that workers' outside options matter for shareholders. Industries with mobile labor expose equity owners to more systematic risk, and the stock market prices that risk.